

Ali Mohamed Abdelhamid

Nasr City, Cairo

+20 100 543 1704 — ali.mohamed.abdelhamid.443@gmail.com — [Linkedin.com/in/alimohamed](https://www.linkedin.com/in/alimohamed)

SUMMARY

Undergraduate Mechatronics Engineering student with practical experience in industrial automation, embedded systems, and CNC manufacturing. Proficient in PLC programming, CAD/CAM, and mechanical design, with a strong focus on integrating mechanical systems with control and embedded solutions to develop efficient and reliable engineering systems.

EDUCATION

Faculty of Engineering, Ain Shams University — Mechatronics Engineering Sep. 2022 – Present
Bachelor of Mechatronics Engineering
GPA: 3.38 / 4.0 (B+)

INTERNSHIP

GS Engi Tech Internship Aug. 2025 – Sep. 2025
Manufacturing Trainee Ludhiana, India

- Designed and prototyped a gaming screen holder for a racing simulator seat (TRONE) using SOLIDWORKS.
- Developed an adjustable mounting mechanism with a secure locking system to ensure stability and usability.
- Optimized the design to meet strict dimensional constraints and improve user flexibility.

PROJECTS

Modular Mecanum-Wheel Mobile Manipulator Jun. 2026

- Designed the mechanical CAD and executed the system integration for a modular mobile manipulator.
- Engineered a decentralized Master-Slave control architecture using ESP32 over ESP-NOW, designing custom PCBs and programming the Inverse Kinematics solver.
- Developed an autonomous navigation state machine and a Web-Based HMI featuring a real-time 3D Digital Twin for seamless manual-to-autonomous teleoperation.

Air-Powered Vehicle Project — Machine Design Jun. 2026

- Designed and built a 2.7 kg compressed-air vehicle achieving 4th place out of 64 teams with a top speed of 2.7 m/s.
- Engineered propulsion and pressure systems, including a 5 mm nozzle and 10 bar tank using Thick Cylinder Theory.
- Developed an ESP32-based wireless braking system with real-time mobile control for precise stopping.

Automated Sliding Door System — Industrial Automation Nov. 2025

- Designed an automated sliding door system using hardwired ladder logic and motor control.
- Built the electrical control panel with relays, contactors, breakers, and safety systems.
- Created 3D panel models using SOLIDWORKS and validated logic using CADE SIMU.

Automated Bottling & Sorting Line — Industrial Automation Dec. 2025

- Designed a full production line simulation using Factory I/O for filling and height-based sorting.
- Programmed PLC logic using Siemens TIA Portal with automated tank and sensor control.
- Developed HMI interface for real-time monitoring, alarms, and process management.

EXPERIENCE

Microsoft Machine Learning Engineer Track (DEPI) Jul. 2026 – Present

- Studying Machine Learning, Deep Learning, Computer Vision, NLP, Azure AI, and MLOps using Python.
- Applying machine learning concepts to engineering and robotics applications through hands-on labs.
- Developing practical AI solutions as part of a capstone project.

NTI Training Program — Embedded Systems Intern Aug. 2025 – Sep. 2025

- Developed embedded drivers for GPIO, ADC, Timers, and Sensors using Embedded C.
- Implemented UART, SPI, and I2C communication protocols on AVR architecture.

- Validated system functionality using Proteus simulations and virtual labs.

Innovation Hub (IHub) — CNC Machining and Laser Cutting Intern

Apr. 2025 – May. 2025

- Learned CNC machine setup, zero-point calibration, and G-code execution.
- Designed and manufactured a 3-pyramid model using Fusion 360 and CNC machining.
- Operated laser cutting systems for precise fabrication and production.

CERTIFICATIONS

• Certified SOLIDWORKS Professional (CSWP) — Dassault Systèmes	Sep. 2026
• Certified SOLIDWORKS Associate (CSWA) — Dassault Systèmes	Dec. 2025
• Building LLM Applications with Prompt Engineering — NVIDIA	Sep. 2025
• Professional PCB Design with Altium — Udemy	Jul. 2025
• Microcontroller-Based Systems — NTI	Sep. 2025

TECHNICAL SKILLS

• CAD / Design:	SOLIDWORKS, Fusion 360, Autodesk Inventor, AutoCAD
• Automation:	Siemens TIA Portal (PLC/HMI), Classic Control
• Embedded Systems:	Embedded C, AVR Architecture, ESP32, Arduino, Proteus
• Simulation:	SolidWorks Simulation, ANSYS, MATLAB Simulink, Factory I/O
• Programming:	C, C++ , Python, MATLAB, SQL
• PCB Design:	Altium Designer, Fusion 360 Electronics